Classifying Students' Meta-Cognitive Comments

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Abstract. We report progress on automatically classifying written comments that students provide after receiving their performance on tests. The aim of this classification is to help teachers support the development of students' metacognitive skills more effectively. We describe a classification pipeline that seamlessly integrates large or small language models (LLMs or SLMs), leveraging state-of-the-art retrieval augmented generation, and human feedback. We apply our approach to field data from high school physics tests and to a classification scheme derived from a model for self-regulated learning. The best classification accuracies achieved for SLMs are of the order of 0.8, which is comparable to what can be obtained with LLMs. The classification obtained indicates that students in similar classroom contexts have very different perceptions and levels of analysis of their performance on assessments. While some focus solely on the factual interpretation of their quantitative results, others comment on their level of confidence, self-efficacy and learning strategies.

Keywords: Self-Regulated Learning · Meta-Cognition · Assessment with confidence levels · Text classification · Generative AI.

1 Context, Motivation and Aims

Leverage the meta-cognitive potential of closed-ended question tests. Closed-ended-question tests in the form of true-false (TFQs) or multiple-choice questions (MCQs) are frequently used in contemporary education due to their ease of use. However, while these tests can indicate whether a student response is correct or incorrect, they fail to offer comprehensive insights into the student's thought processes. Their potential to develop students' meta-cognitive skills is under exploited. To bridge this gap, a first step is to ask students to provide the level of confidence in their answer to a test [4, 5]. In this way, students can express their doubts and obtain more precise feedback, combining level of confidence and correctness of their answers [2]. To go a step further, Hoffmann *et al.* [9] requested meta-cognitive comments from students after test completion. This encourages students to reflect on their reasoning and to comment the test feedback received. Example comments include *C1: "Mechanics is a difficult subject, and I'm generally overconfident"*, or *C2: "I could be more realistic if I read the questions better"*. Analyzing this data could reveal patterns in student thinking, highlight misconceptions, and provide levers to support meta-cognitive activity.

Objective. Our work aims to support teachers by automating the classification of students' meta-cognitive reflections on their knowledge and errors as observed in their test results. We want to classify those comments into relevant categories based on a model for self-regulated learning (SRL) [15].

Challenges. Our work faces several challenges. First, the students' comments are in part subjective perceptions of their results, expressed using varied language, tone, and style, and can therefore be ambiguous. Second, the meaning of subjective data is often highly context-dependent, in our case, the test subject, the test environment, and the form of feedback delivered. This contextual nature means that the specific situation must be taken into account to classify the data. Generative AI based on language models (LMs) is a good candidate for achieving this. However, here we face a third challenge: as is often the case in educational contexts, our dataset has a limited amount of data, making it impossible to train the LM for our particular task using a supervised learning method. The final challenge concerns the definition of a ground truth. More than one category may be suitable for a ambiguous comment (see example in section 4), which complicates the validation of our process.

Research questions. We formulate three research questions. *RQ1: How can we generate and validate a classification scheme for meta-cognitive comments using LMs? ; RQ2: What classification accuracies can we achieve? ; RQ3: What kind of information can the teacher obtain from such a classification?*

2 Conceptual and Technical Background

Metacognition and SRL. According to Flavell's original work [8], metacognition is the awareness and regulation of an individual's cognitive activities in learning processes. As early as the 1990s, a meta-analysis by Wang et al. [14] revealed that meta-cognitive skills have a significant positive effect on academic performance. Moreover, according to the social-cognitive perspective [16], metacognition is also an essential component in learners' self-regulation. While it is important for all students to develop self-regulation skills, only some learners achieve this spontaneously, drawing inspiration from their parents, teachers or peers. Others do not, and so it becomes necessary to stimulate and support them [13].

Subjective data classification with LMs. Prior work [9] testing lexical analysis to classify students' comments did not yield satisfactory results. In the present study, we explore the use of LMs. A priori, they offer advanced capabilities in capturing nuanced contextual information and the variability of human language [12]. However, the amount of data to be provided to the LM to adapt its general capabilities to the specific context is too small for supervised learning and too large to be included in the prompt's limited input window. The RAG (Retrieval-Augmented Generation) technique is a promising solution to this problem [7]. Depending on the specific query formulated, RAG provides the LM with relevant contextual information, extracted from an external, previously customized knowledge database.

Need for Human-in-the-loop. Shah *et al.* [11] combine LLMs with humans to generate and validate taxonomies to analyze user intent in logs. In our study, the categories are derived from a founded SRL model, offering two benefits: the possibility of model validation and theory building, and secondly, the identification of areas of meta-cognitive activity that are neglected by students. Indeed, if the LM is used to select categories, certain aspects will not be apparent to the teacher. We have chosen the COPES model for self-regulated learning [15] because it is characterized by a marked meta-cognitive perspective. Nevertheless, the teacher has to adapt the categories to the teaching context and therefore be involved in defining and validating the classification scheme.

3 Data and Measures

Data. We collected two datasets from high school students, aged 17 or 18. They participated in physics tests, each composed of about twenty TFQs. For each question, students were asked to indicate their degree of certainty that their answer was correct, out of 50%, 60%, 70%, 80%, 90% or 100%. At the end of the test, they were given a summary of their results incorporating accuracy and degrees of certainty, and were invited to provide written metacognitive comments on their results, errors and possible remedies. Corpus 1 was produced by 178 students in OCT/NOV 2019. Their comments, which generally touch on several meta-cognitive aspects, were segmented into 602 units of meaning and classified in a previous study [9]. Three months later, 105 of them took a second test, producing Corpus 2 with 235 meaning units. In what follows, we use the term "student comments" to refer to units of meaning, each addressing a single meta-cognitive aspect.

Measures of classification quality. We use Cohen's Kappa, a statistical measure used to evaluate the level of agreement between two coders (LM and Human or Human and Human) taking into account what would be expected by chance [3]. We also use accuracy to measure the ability of a RAG-LM system to assign comments to the correct category with regard to a reference classification. It is defined as the number of correctly classified comments divided by the total number of comments.

4 Contributions

Generation and validation of a classification scheme for meta-cognitive comments. We now address RQ1 and RQ2. The first step in using a RAG is to supply useful information to a vector database (DB), ChromaDB in our case. The data of Corpus 1 was split into two parts: 2/3 of the comments and their categories were embedded as Base Data, and the remaining 1/3 was used as Refine Data. The latter comments were classified using RAG and GPT3.5 as LLM, see Fig.1a. This classification was compared to the human reference classification established in the previous study [9]. Three actions allowed us to significantly improve classification accuracy (roughly from 0.45 to 0.8), as part of an iterative

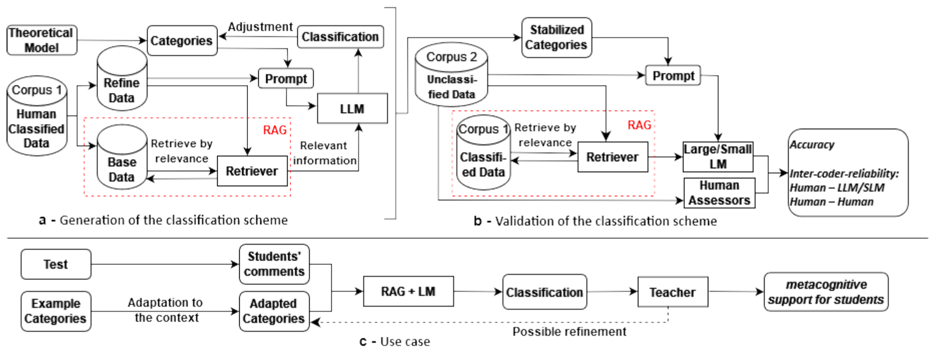

Fig. 1. Generation, validation and use of a classification scheme for meta-cognitive comments.

process: a) Moving the category descriptions, initially embedded together with the Base Data in the DB, to the LM prompt, making it easy to adjust these descriptions without modifying the DB; b) Adding representative keywords in the category description; c) Reducing the number of categories: starting with 13 categories, we merged some of them and redefined the boundaries of others which yields 9 stabilized categories, representing three different types of meta-cognitive processes in the COPES model [15]: cognitive evaluations of the test results (3 categories), updates of learner's cognitive conditions (5 categories), and forward planning, i.e., projection into future tasks [1]. This human-in-the-loop refinement process ensured that the resulting categories were both suitable for automated classification and meaningful for the analysis of students' meta-cognition.

In a second phase, we validated the obtained classification, using Corpus 2, see Fig. 1b. Corpus 1 was now fully supplied to the DB. To establish a ground truth for LM's categorization evaluation, two researchers manually classified the 235 comments separately using the 9 stabilized categories. Disagreement arose over certain comments. While comment *C1* provided in Section 1 can be classified as an update of the student's self-efficacy, comment *C2* is more ambiguous as it may reflect an update of the student's knowledge of his study tactics (taking more time to study the questions posed) but also a projection into future tasks. The inter-coder-reliability measured using Cohen's Kappa was 0.67 what indicates substantial agreement (0.61 to 0.8) [1]. In a second step, discrepancies were discussed by the two coders until an agreement was found, constituting the reference to calculate accuracy and Cohen's Kappa for LM classifications. We compare in Table 1 the results achieved with three different LMs: Llama 3.3, Phi-4, and Mistral 7B, using 3 GPU NVIDIA Quadro RTX 8000 48 GB.

We can see that the LM Phi-4 offers the best trade-off between the classification quality and the required energy consumption/runtime. The accuracy of 0.8 is comparable to other work [10] and acceptable given the subjective nature of the data. Cohen's Kappa of 0.77 is better than the value initially obtained between the two human coders and similar to prior work [11]. As far as disagreements are concerned, we find discrepancies between LM and Human classifications similar to those between the two human coders [1]. Finally, to assess the reproducibility of our process, we repeated the classification with each LM 3 times. Acceptable variations in accuracy between 0.02 and 0.04 were observed.

Insights in students' meta-cognitive capacities. We now turn to RQ3 and what teachers can learn from the comments. Analysis of Corpus 2 reveals that 3/4 of students factually discuss their results in terms of correctness and confidence level; 1/3 identify disciplinary gaps; around 20% report feedback on their self-efficacy: confirmation or adjustment of their perceived competence; 40% question their learning strategies and 20% address forward planning. Compared with the first test and Corpus 1, we can observe that more students comment on their degree of certainty, likely because they are getting used to this new type of assessment. Comments on learning strategies and the impact on future study tasks are slightly more present, which is a positive evolution. However, as before, very few students (3%) comment on motivational and emotional aspects.

5 Transfer in Practice and Future Research Agenda

The proposed classification process could be transferred to the practice as shown in Fig. 1c, but the following points would need to be addressed upstream: a) constitution of a larger database with classified meta-cognitive comments for RAG; b) automated division of long comments into units of meaning; c) constitution of a larger set of examples of relevant categories, always with regard to the theoretical model. Teachers could then start from the example categories and adjust them: delete or add categories, choose keywords that are representative of their context, i.e. that characterize their teaching subject, but also the specific vocabulary that the learning platform uses or that they have established with students. A SLM and a database for RAG could be installed on the learning platform's server, and classification carried out locally, thus respecting data confidentiality. After an initial run, teachers must evaluate the quality of the obtained classification and, if necessary, iteratively adjust the categories and keywords.

To move in this direction, we plan to : (1) Apply our process to comments from a variety of contexts to test it and enrich Corpus 1 to feed the database; (2) Test multi-LM-agent configurations using state-of-the-art [6], where each LM handles a specific function (e.g., segmentation of long comments, classification, evaluation of classification quality).

LM	**Accuracy**	**Cohen's Kappa**	**Energy consumption**	**Runtime**
Mistral (7B)	0.69	0.66	6.6 Wh	1m31s
Phi-4 (14B)	0.80	0.77	14.7 Wh	2m42s
LLaMA 3.3 (70B)	0.80	0.76	42.5 Wh	8m34s

Table 1. Accuracy, Cohen's Kappa, Energy consumption, and Runtime for the classification of 235 student comments with three different LMs.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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